Do Cointegration-based Pairs Remain Profitable?

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Cointegration-based pairs

- A pair trade has a formation step that selects assets and specifies spreads, and a trading step that opens positions upon divergence and closes them upon mean reversion.
- Cointegration (Engle and Granger, 1987) is the standard formation tool, extended to baskets by Johansen (1988) and principal components (PCs) of Stock and Watson (1988).
- However, prior empirical literature documents declining profitability. Net returns were -2.76% a year in 2010 to 2015 (Clegg and Krauss, 2018), and -5.49% cumulative for distance-selected pairs in 2020 to 2023 (Reichmann, 2026).
- Cointegration found at formation rarely persists. The share of pairs that remain cointegrated a year later is about the size of the test (Clegg, 2014). Truly cointegrated stocks would imply Sharpe ratios above 10, far beyond those observed (Farago and Hjalmarsson, 2019).

Research question

Question

1. Is cointegration-based formation profitable out of sample across combination methods, stock groupings and window lengths?
2. Does formation-stage mean reversion persist?

Finding

1. No formation choice can consistently generate profit on its own. In the testing period from 2020 to 2025, the top decile of every method loses between 0.70% and 1.34% a year.
2. The formation p-value does not consistently rank returns, and neither grouping nor window changes the sign of the result.
3. Formation-stage mean reversion persists little more than that of random walk spreads.

The data are daily prices of point-in-time S&P 500 constituents. Trading windows starting in 2010 to 2019 form the training sample, and those in 2020 to 2025 the testing sample.

Spreads and grouping

$$s_t = \sum_{i=1}^{m} w_i \, \tilde{p}_{i,t}, \qquad \|\mathbf{w}\|_1 = 1, \qquad \tilde{p}_{i,t} = P_{i,t}/P_{i,t_0}$$

- $P_{i,t}$ is the adjusted closing price of stock i on day t, and t_0 the last day of the formation window Φ, the $L \in \{1, \ldots, 5\}$ years before a formation date.
- The weights of a spread of m stocks are estimated on Φ and held fixed over the trading window E of $H \in \{1, 3, 6, 12\}$ months.
- Sectors Each of the 11 sectors is one group.
- OPTICS clusters Within each sector, stocks are clustered with OPTICS on their PC loadings above the noise bound of random matrix theory, after removing the market mode.
- Complete-linkage clusters Within each sector, the returns are clustered by complete linkage on a correlation distance, with as many clusters as PCs above the noise bound.

Combination methods

Method	Weights	Formation statistic
Pairs	Regression of one normalized price on the other	Smaller ADF τ
Johansen	First cointegrating vector (Johansen, 1988)	Trace statistic
PCA	Smallest PC of normalized price levels (Stock and Watson, 1988)	ADF τ of the spread
Eigenanalysis	Smallest eigenvector of squared autocovariances sum (Zhang et al., 2019)	ADF τ of the spread
Stock versus group	Regression on the equal-weighted rest of its group	ADF τ of the residual

Combination methods using the augmented Dickey–Fuller (ADF) statistic τ.
All methods except pairs require a group of $m \geq 3$ stocks.

Random walk null

$$dX_{i,t} = \sigma_i \, dW_{i,t}, \qquad i = 1, \ldots, m$$

- Standard Dickey–Fuller critical values cannot be applied to the formation statistics, because each method chooses its weights to make the spread look stationary.
- In pairs, as in Engle and Granger (1987) and MacKinnon (2010), critical values must be simulated on independent random walks to correct for selection bias.
- We therefore apply each method to independent random walks, with log prices $X_{i,t}$ driven by independent standard Brownian motions $W_{i,t}$.
- Each σ_i is drawn from the daily return standard deviations from 2005 to 2019 of all stocks with usable prices, and the daily paths have length $T = 252L$.

Formation p-values

$$d = \min\{\lfloor 10p \rfloor, 9\}, \qquad \hat{\pi}_0 = \min\{1, 2\,\#\{p \geq 0.5\}/N\}$$

- The p-value p of a spread is the share of null statistics at least as extreme as its own.
- Spreads are assigned to deciles d, where $d = 0$ ($d = 9$) denotes the strongest (weakest) formation evidence.
- $\hat{\pi}_0$ estimates the proportion of true nulls among N spreads (Storey, 2002).

Trading rule

$$n_t = \begin{cases} -\operatorname{sgn}(z_t) & \text{if } z_t\, n_{t-1} \geq 0 \text{ and } |z_t| \geq 2, \text{ and } t \text{ is not the last day of } E, \\ 0 & \text{if } n_{t-1} \neq 0,\ z_t\, n_{t-1} \geq 0 \text{ and } |z_t| < 2, \text{ or } t \text{ is the last day of } E, \\ n_{t-1} & \text{otherwise.} \end{cases}$$

- $n_t \in \{-1, 0, 1\}$ is the position set at the close of day t by the rule of Gatev et al. (2006), with $z_t = (s_t - \bar{s}_\Phi)/\hat{\sigma}_\Phi$. $\bar{s}_\Phi$ and $\hat{\sigma}_\Phi$ are the sample mean and standard deviation.
- A position opens when the spread is 2 standard deviations from its mean, and closes when the spread returns to the mean or the window ends.
- The net return r of a spread is its gain over E on one unit of capital after costs.
- Trading costs 10 basis points (bps) per unit traded, and borrowing costs 25 bps a year on the legs held short.

Measures and portfolios

$$r_k = \frac{1}{|\mathcal{C}_k|} \sum_{(g,L)\in\mathcal{C}_k} \bar{r}_{g,L,k}$$

- For each method, $\bar{r}_{g,L,k}$ is the mean net return of the top-decile spreads of grouping g and window L in trading period k.
- $\mathcal{C}_k$ is the set of (g, L) that hold at least one top-decile spread in period k, and r_k is the method portfolio return.
- Persistence is measured by the ADF statistic for the spread over the formation and trading windows, written τ_Φ and τ_E.
- The re-test passes when τ_E lies below the 5% critical value of MacKinnon (2010).

Formation evidence and persistence

			τ_Φ		τ_E		Re-test pass (%)	
Method	$\hat{\pi}_0$	Spreads	Data	Null	Data	Null	Data	Null
Training, 2010 to 2019								
Pairs	1.00	6014	−3.77	−3.78	−1.57	−1.51	5.7	4.8
Johansen	0.70	84	−7.70	−7.17	−1.61	−1.51	5.9	4.8
PCA	0.66	84	−7.78	−7.64	−1.66	−1.53	7.2	5.2
Eigenanalysis	0.68	83	−7.90	−7.75	−1.65	−1.50	6.6	4.7
Stock versus group	0.95	592	−3.58	−3.57	−1.54	−1.51	5.0	5.1
Testing, 2020 to 2025								
Pairs	1.00	7579	−3.75	−3.78	−1.63	−1.51	5.5	4.8
Johansen	0.68	116	−8.25	−7.63	−1.69	−1.51	7.2	4.8
PCA	0.76	80	−7.80	−7.65	−1.75	−1.52	8.9	5.1
Eigenanalysis	0.77	84	−8.23	−8.07	−1.75	−1.49	9.0	4.7
Stock versus group	1.00	606	−3.56	−3.58	−1.62	−1.51	5.7	5.1

Formation evidence and persistence, $H = 6$ months. Spreads is the mean number of top-decile spreads per trading period, and the τ and re-test columns are for the top decile. $\hat{\pi}_0$ is computed over all spreads.

- With $\hat{\pi}_0$ of 0.95 to 1.00, formation evidence for pairs and stock-versus-group spreads is virtually indistinguishable from pure chance, against 0.66 to 0.77 for baskets.
- Out of sample, the selected top-decile spreads decay rapidly toward null behavior.

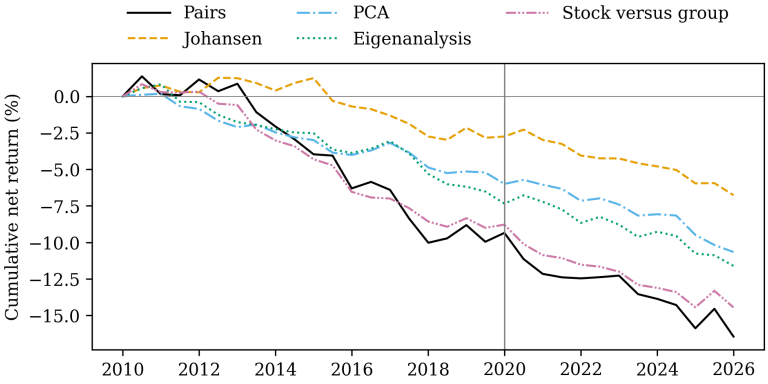
Net returns

	Net return		Null	Lift		Top minus bottom	
Method	Mean	t	Mean	Mean	t	Mean	t
Training, 2010 to 2019							
Pairs	−0.97	−2.27	−1.12	0.15	0.35	0.27	0.43
Johansen	−0.28	−1.17	−1.07	0.79	3.13	1.67	3.56
PCA	−0.62	−3.58	−1.42	0.81	4.58	−0.40	−0.81
Eigenanalysis	−0.76	−3.24	−1.57	0.81	3.49	0.43	0.65
Stock versus group	−0.91	−3.20	−1.12	0.21	0.71	1.70	2.68
Testing, 2020 to 2025							
Pairs	−1.34	−3.07	−1.12	−0.23	−0.52	−0.14	−0.14
Johansen	−0.70	−3.89	−1.11	0.41	2.46	−0.07	−0.18
PCA	−0.84	−3.62	−1.46	0.61	2.39	0.20	0.38
Eigenanalysis	−0.78	−3.60	−1.70	0.91	3.51	0.59	0.86
Stock versus group	−1.06	−3.93	−1.14	0.07	0.27	2.07	1.48

Annualized net return (%) of the top-decile portfolio, $H = 6$ months. Lift is the net return minus the null.

- Every method loses, by a significant 0.70% to 1.34% a year in the testing sample.
- Baskets earn a significant lift of 0.4% to 0.9% a year over the null.
- No top-minus-bottom return is significant in the testing sample.

Cumulative net return



Cumulative net return of the top-decile portfolio, $H = 6$ months. The vertical line marks the start of the testing sample.

- Losses accumulate steadily rather than stemming from localized tail events. The first half of 2020, which includes the market crash, loses 0.41% on average across the 5 methods.

Formation windows and trading horizons

	Training, 2010 to 2019				Testing, 2020 to 2025			
L (years)	$H = 1$	$H = 3$	$H = 6$	$H = 12$	$H = 1$	$H = 3$	$H = 6$	$H = 12$
1	−1.79	−1.14	−1.12	−1.22	−1.49	−1.11	−0.48	−0.89
2	−1.15	−0.59	−0.54	−1.46	−1.49	−1.29	−0.92	−1.33
3	−1.08	−0.90	−0.92	−0.98	−1.79	−1.39	−1.57	−1.14
4	−1.03	−0.75	−0.55	−0.72	−1.07	−0.90	−1.04	−0.38
5	−1.21	−1.04	−0.67	−0.46	−1.12	−0.85	−0.82	−0.31

Annualized net return (%) of the equal-weighted average of the 5 method portfolios over the three groupings, by formation window L and trading horizon H in months.

- All 40 averages are negative and mostly significant, and 1-month horizons incur the steepest losses on average.
- Relative performance across formation windows shows no stable pattern across horizons or sample periods.

Grouping

- Grouping choice similarly fails to achieve profitability, though cluster-based spreads lose less than industry sectors.
- With $H = 6$ months, sectors lose 0.86% a year in training and 1.33% in testing, against 0.50% and 0.60% for OPTICS and 0.73% and 0.98% for complete linkage.
- OPTICS clusters significantly outperform sectors by 0.73% a year out of sample, whereas complete-linkage clusters yield a smaller, statistically insignificant advantage.

Conclusion

- We form S&P 500 spreads by 5 cointegration-based methods within sectors and correlation-based clusters, and trade them by a 2-standard-deviation threshold rule.
- From 2020 to 2025, the top decile of every method has net losses of 0.70% to 1.34% a year, which persist across windows, groupings, exit levels and costs (Appendix B.1).
- Formation p-values fail to consistently rank returns, and out-of-sample mean reversion persists only marginally beyond random walk expectations.
- The positive lift of baskets over the random walk null likely reflects uncaptured common factors or volatility clustering rather than true cointegration.

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Code and replication data are available at *GitHub*.

Appendix B.1. Exit levels and costs

Exit at $\lvert z_t \rvert$	Cost (bps)	Training		Testing	
		Mean	t	Mean	t
0.5 before the mean	5	−0.38	−2.02	−0.56	−3.76
	10	−0.76	−3.93	−0.96	−6.45
	25	−1.91	−8.80	−2.18	−12.64
0 at the mean	5	−0.36	−1.89	−0.58	−4.44
	10	−0.71	−3.62	−0.95	−7.04
	25	−1.75	−8.20	−2.04	−12.61
0.5 past the mean	5	−0.37	−1.89	−0.55	−4.53
	10	−0.69	−3.47	−0.89	−7.09
	25	−1.66	−7.77	−1.91	−12.58

Annualized net return (%) of the equal-weighted average of the 5 method portfolios under alternative exit levels and costs, $H = 6$ months.

- The exit condition $z_t\, n_{t-1} \geq 0$ is replaced by $z_t\, n_{t-1} \geq -0.5$ and $z_t\, n_{t-1} \geq 0.5$.
- Relative to the exit at the mean, the exit level changes the return by at most 0.17% a year, and every variant loses money in both samples.